UNIVERSITI MALAYA

WID3010 Autonomous Robots
Semester 2, 2024/2025
Case Study

Occurrence: 1

Group: 1

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System Analysis: Critical needs of the system

1. **Real-Time Detection (20 FPS)**
   The system achieves real-time performance through a dedicated ***groove_detector_node***, which runs independently and leverages GPU acceleration to process visual data. To maintain efficiency and minimize communication overhead, the node transmits lightweight ROS messages (***geometry_msgs/Point***) representing groove center coordinates, rather than large image data.

2. **3D Position Calculation**
   The ***position_calculator_node*** performs the critical task of converting 2D pixel coordinates from the YOLOv5 output into 3D space by fusing them with depth information from the RealSense camera. This process relies on pre-calibrated camera intrinsic parameters, as detailed in Table 1 of the source paper, ensuring accurate projection and spatial mapping.

3. **Hand-Eye Coordination**
   To enable precise robot movement, the ***transform_manager_node*** applies hand-eye calibration using a precomputed rotation matrix RRR and translation vector TTT, as outlined in Equations 8 and 9 of the study. These static transformations are broadcasted via TF, enabling low-latency, real-time conversions between the camera frame and the robot's Tool Center Point (TCP) coordinate system.

4. **Socket Communication with Robot**
   The ***fanuc_bridge_node*** facilitates direct communication between the ROS system and the Fanuc robot by handling protocol translation. As described in Section 4.2 of the referenced paper, this node uses custom TCP/UDP socket protocols instead of relying on ROS-Industrial, allowing tighter integration with the specific hardware setup used in the experiment.

5. **Multi-Distance Operation (0.3–2 m)**
   The system supports robust operation over a wide range of distances (0.3 to 2 meters), with depth data processed directly through the native RealSense SDK. Notably, no adaptive parameters are needed to handle different distances, as the paper reports a consistently low error margin of less than or equal to 2% across all tested distances.

ROS Architecture: Nodes, topics, publishers and subscribers

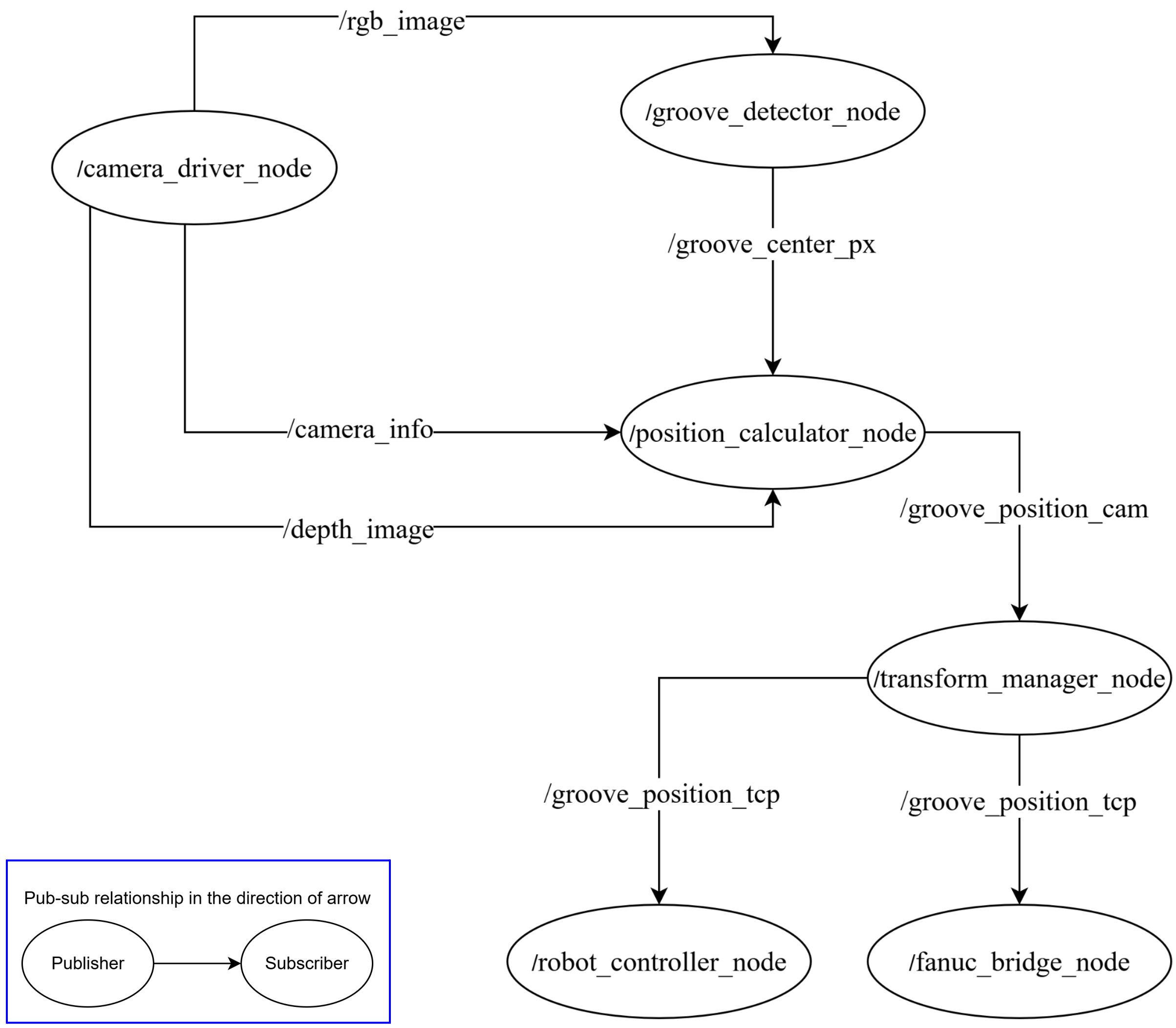
Node Name	Purpose	Publisher	Subscriber	Key Detail
camera_driver_node	Interfaces with RealSense D435i depth camera	/rgb_image (sensor_msgs/Image): Raw RGB images at 1280×720 resolution. /depth_image (sensor_msgs/Image): Depth maps aligned with RGB data. /camera_info (sensor_msgs/CameraInfo): Intrinsic parameters (Table 1: f_x, f_y, u_0, v_0)	None	Uses librealsense ROS wrapper. No IR projector control (paper uses fixed settings)
groove_detector_node	Runs improved YOLOv5s + CA model for weld groove detection	/groove_center_px (geometry_msgs/Point): Pixel coordinates (x_{cen}, y_{cen}) of groove center (Eq. 10).	/rgb_image: Input for YOLOv5 inference.	Outputs only center coordinates (not bounding boxes). Optimized for 20 FPS on GPU (NVIDIA GTX 960).
position_calculator_node	Computes 3D groove position using depth data	/groove_position_cam (geometry_msgs/PointStamped): 3D position (x, y, z) in camera frame (Fig. 10).	/groove_center_px: Groove center from detector. /depth_image: Depth data for 3D calculation. /camera_info: Camera intrinsics for projection (Eq. 7).	Uses OpenCV to convert pixels to 3D points.

transform_manager_node	Applies hand-eye calibration to convert camera coordinates to robot TCP frame	/tf (tf2_msgs/TFMessage): Static transform from camera to TCP (Eq. 8 rotation R, Eq. 9 translation T). /groove_position_tcp (geometry_msgs/PointStamped): Transformed position in TCP frame.	/groove_position_cam: Position in camera frame.	Pre-calibrated transform (no online recalibration per paper).
robot_controller_node	Guides robot to target position via socket communication	None	/groove_position_tcp: Target position in TCP frame.	Implements socket communication with Fanuc control cabinet (paper Section "Experiment on weld groove identification and guiding"). No error recovery logic (paper only reports errors in Fig. 15-16).
fanuc_bridge_node	Translates ROS messages to Fanuc socket protocol	None	/groove_position_tcp: Receives target positions.	Uses TCP/UDP sockets as specified in paper (Page 11).

Topic Name	**Message Type**	**Purpose**
/rgb_image	sensor_msgs/Image	Raw RGB input for weld groove detection.
/depth_image	sensor_msgs/Image	Depth data for 3D position calculation.
/camera_info	sensor_msgs/CameraInfo	Camera intrinsics for coordinate projection.
/groove_center_px	geometry_msgs/Point	Pixel coordinates (x_{cen}, y_{cen}) of groove center.

/groove_position_cam	geometry_msgs/PointStamped	3D groove position in camera frame.
/groove_position_tcp	geometry_msgs/PointStamped	Transformed position in robot TCP frame.

RQT Graph



<u>Note:</u>
One of the transform manager node's topic (published via **/tf**) is not explicitly shown as subscribed topics in the RQT graph because:

- Special Handling: The TF system operates behind the scenes via tf2_ros, where nodes implicitly access transforms using TransformListener without direct topic subscription.

- Static vs. Dynamic: Since this system uses a pre-calibrated static transform (camera → TCP from Eq. 8-9), it is broadcast once and cached by the ROS tf2 system.

References

geometry_msgs - ROS Wiki. (2024). Ros.org. https://wiki.ros.org/geometry_msgs

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Li, M., Huang, J., Xue, L. et al. A guidance system for robotic welding based on an improved YOLOv5 algorithm with a RealSense depth camera. Sci Rep 13, 21299 (2023). https://doi.org/10.1038/s41598-023-48318-8

sensor_msgs - ROS Wiki. (2022). https://wiki.ros.org/sensor_msgs

tf2 - ROS Wiki. (2019). Ros.org. https://wiki.ros.org/tf2